

See Youtube Video Here: <https://youtu.be/t-wlqmm4lrw>

[How Can Open Source Engineering Help Regular people?]

Cost : material total is reduced

skill : alignment challenge is overcome

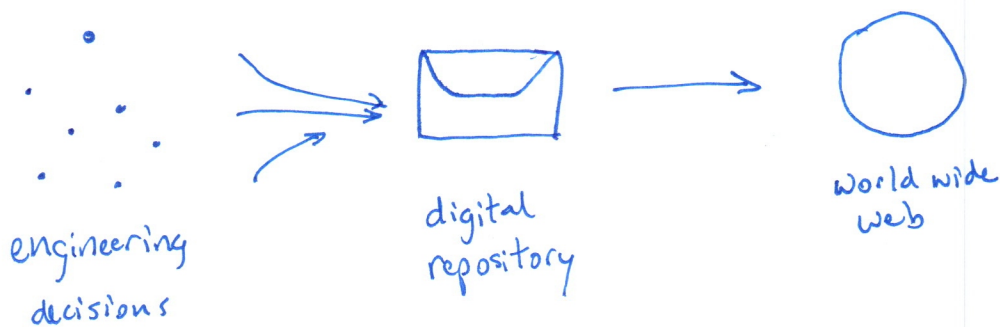
material selection : requires experience : poor choice will require a plane (\$300)

strength: integrate strength of fiberglass, eliminate fixturing & gluing (corner brackets \$30)

----- above this line -----

we answered: how does engineering design improve a product?

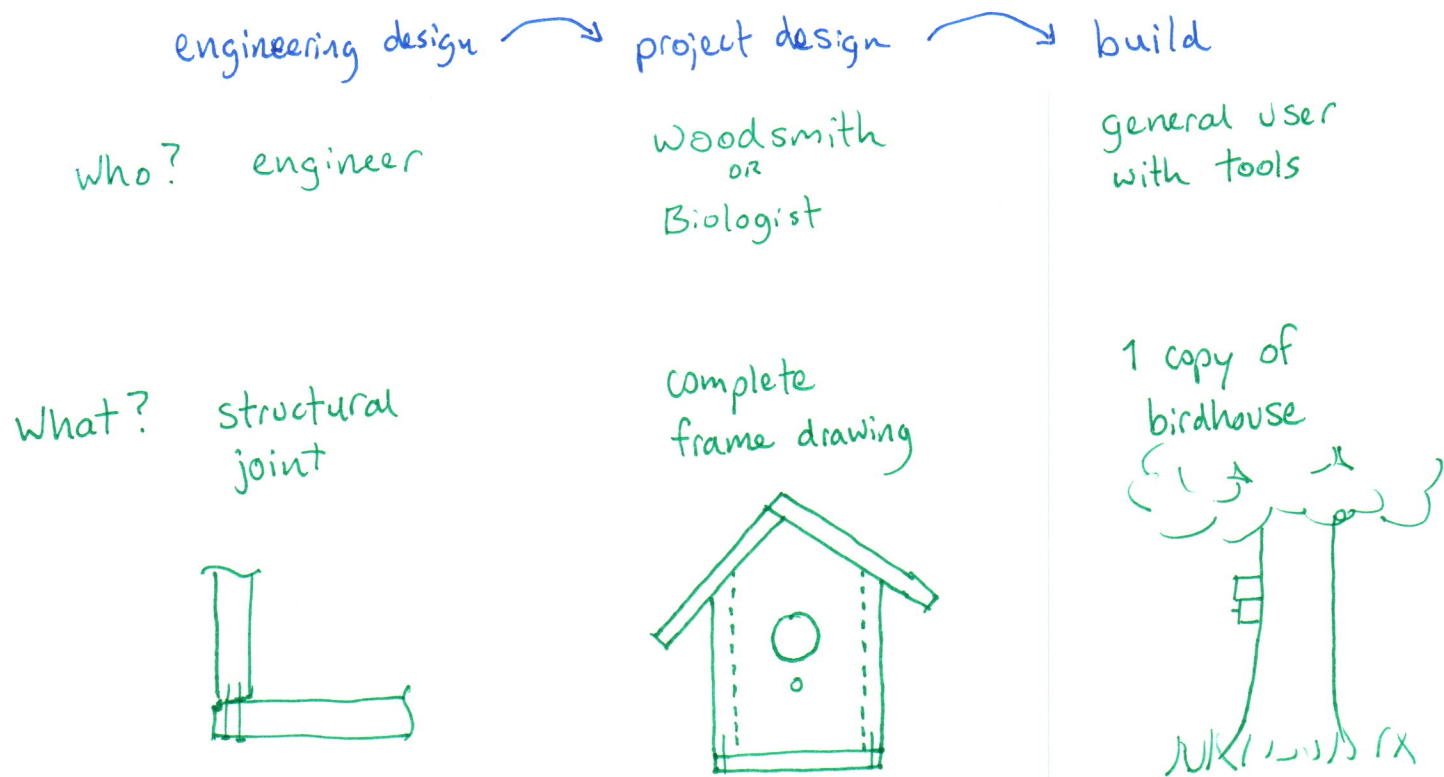
benefits must spread w/zero effort, unlimited no. times



----- above this line -----

You might mistake this process for a "free design plans" actually, "free build plans."

The difference (open source design Versus FREE build plans)



Final Answer to Today's Question

- In an Infinite number of ways
↳ if we collaborate
- In zero ways
↳ if we don't

(Example of a Free design)

Two-chamber Rocket Box

Materials (makes one house)

2" diameter (2 3/8" outside diameter) steel pole, 20' long

Two 1" x 4" (3/4" x 3 1/2" finished) x 8' boards*

Two 1" x 8" (3/4" x 7 1/4" finished) x 8' boards*

* Western red cedar
or poplar preferred

Two 1" x 10" (3/4" x 9 1/4" finished) x 6' boards*

24" x 24" x 3/4" piece of AC exterior plywood

Box of 100 exterior-grade screws, 1 1/2"

Box of 100 exterior-grade screws, 1 1/4"

16 to 32 exterior-grade screws, 2"

20 to 30 roofing nails, 3/8"

One quart water-based primer, exterior grade

Two quarts flat, water-based stain or paint,

exterior grade

Asphalt shingles or dark galvanized metal

One tube paintable latex caulk

Two 1/4" x 4 1/2" carriage bolts, washers and nuts

Recommended tools

→ Table saw or circular saw

• Caulk gun

→ Hammer

Tape measure

Square

Jigsaw, keyhole saw or router

Sandpaper or sander

Rasp or wood file

Variable-speed reversing drill

1 1/2" hole saw or spade bit

3/8" and 1/4" drill bits

Screwdriver bit for drill

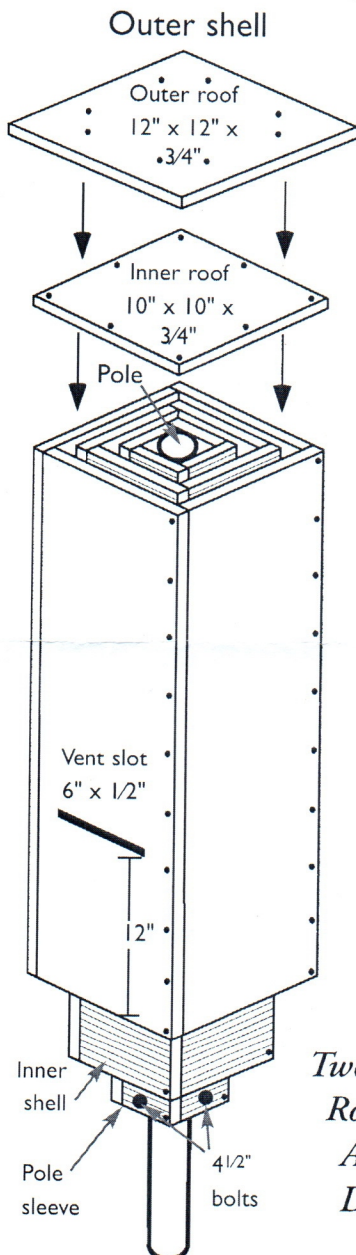
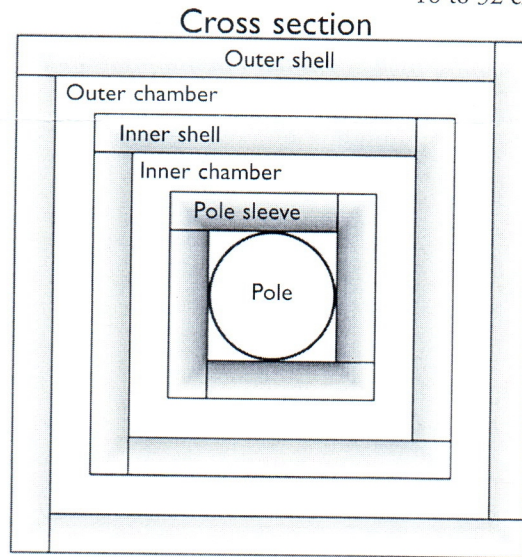
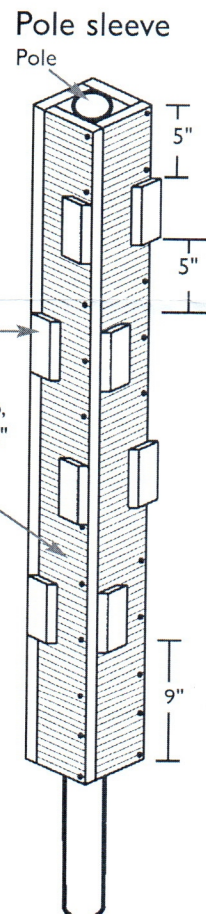
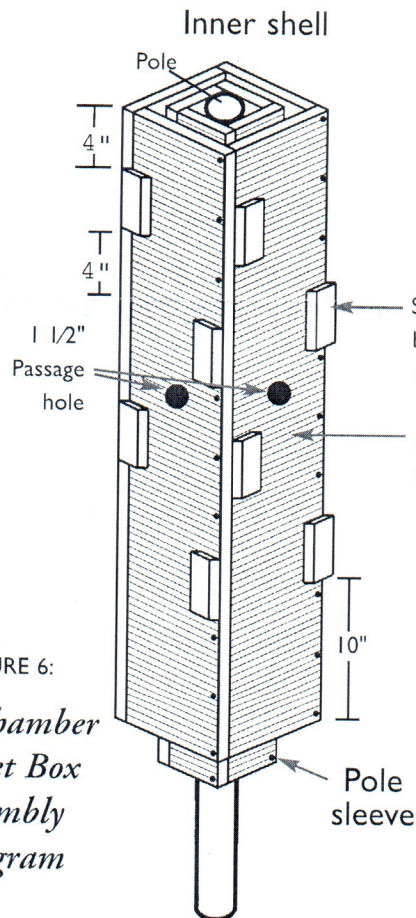


FIGURE 6:

Two-chamber
Rocket Box
Assembly
Diagram



Spacer
blocks
Grooves
1/16" deep,
1/4" to 1/2"
apart

Construction

1. Measure, mark and cut out parts according to Figure 7. Dimensions must be exact for correct fit. Cut out two vent slots and four passage holes as shown.
2. Cut 1/8"-deep horizontal grooves 1/4" to 1/2" apart on one side of all 36" and 45" boards and on both sides of all 42" boards. Sand to remove splinters.
3. Drill two 3/8" holes through each 3/4" x 1 1/2" x 4" spacer block to prevent splitting.
4. Assemble four pole sleeve boards into a hollow, square box as shown using 1 1/2" screws and caulk. Pre-drill holes to prevent splitting. Countersinking holes may also help.